

Energy Domain Challenge (SQL + Power BI)

Problem Statement

You are working as a **Data Analyst in an Energy Company**.

Goal:

Analyze electricity consumption, revenue, and customer behavior to optimize energy distribution and increase profitability.

Step 1: SQL – Database Setup

◆ Create Table

```
CREATE TABLE Energy_Consumption (  
  customer_id INT,  
  region VARCHAR(20),  
  energy_source VARCHAR(20),  
  consumption_kwh FLOAT,  
  billing_amount FLOAT,  
  payment_status VARCHAR(20),  
  bill_date DATE  
);
```

◆ Insert Records

```
INSERT INTO Energy_Consumption VALUES  
(101, 'South', 'Solar', 120, 1500, 'Paid', '2024-01-01'),  
(102, 'North', 'Coal', 300, 3500, 'Pending', '2024-01-02'),  
(103, 'East', 'Wind', 200, 2500, 'Paid', '2024-01-03'),  
(104, 'West', 'Hydro', 400, 4500, 'Paid', '2024-01-04'),  
(105, 'South', 'Coal', 350, 4000, 'Pending', '2024-01-05'),  
(106, 'North', 'Solar', 150, 1800, 'Paid', '2024-01-06'),  
(107, 'East', 'Hydro', 280, 3200, 'Pending', '2024-01-07'),  
(108, 'West', 'Wind', 220, 2600, 'Paid', '2024-01-08'),  
(109, 'South', 'Solar', 180, 2100, 'Paid', '2024-01-09'),  
(110, 'North', 'Coal', 500, 6000, 'Pending', '2024-01-10');
```

Step 2: SQL Tasks

◆ Basic Analysis

-- Total Consumption

-- Total Revenue

◆ **Group By Analysis**

-- Consumption by Region

-- Revenue by Energy Source

◆ **Business Questions**

-- Highest Consuming Region

-- Pending Payments Count

-- Average Billing per Customer

-- Top 3 Customers by Consumption

-- Payment Status Revenue Split

Step 3: Power BI Tasks

◆ **Import Data**

- Connect SQL → Load Energy_Consumption

◆ **Create Measures (DAX)**

Total Consumption

Total Revenue

Pending Payments

Avg Consumption

Step 4: Dashboard Requirements

KPIs

- Total Consumption
- Total Revenue
- Pending Payments
- Avg Consumption

Visuals

- Bar Chart → Consumption by Region

- Column Chart → Revenue by Energy Source
- Pie Chart → Payment Status
- Line Chart → Consumption Trend (bill_date)

Filters

- Region
- Energy Source
- Payment Status

Step 5: Business Insights

1. Which region consumes the most energy?
2. Which energy source generates highest revenue?
3. How much revenue is stuck in pending payments?
4. Which customers should be targeted for payment recovery?
5. Trend: Is consumption increasing or decreasing?